Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
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Dancing with Data: A Mathematical Framework for Fan Vote Estimation and Voting System Optimization in DWTS

Summary

Dancing with the Stars (DWTS) combines professional judge scores with audience votes to determine weekly eliminations, yet the exact fan vote totals remain confidential. This paper develops a comprehensive mathematical framework to estimate fan votes, compare voting methods, analyze influencing factors, and design an improved voting system.

For Problem 1, we formulate fan vote estimation as a constrained optimization problem using a Softmax voting model. Stratified analysis reveals a critical finding: the rank-based method achieves 38.4% consistency while the percentage-based method achieves 0%, suggesting fundamentally different estimation challenges. Bootstrap resampling yields a certainty index of 0.995, demonstrating high estimation stability.

For Problem 2, we apply counterfactual analysis comparing rank-based and percentage-based methods across all 34 seasons. The methods produce identical elimination decisions in 95.5% of cases, with the judge tiebreaker rule changing outcomes in 3.9% of weeks. ANOVA testing shows industry affects judge scores more strongly ($F=8.81$, $p<0.0001$) than fan votes ($F=2.14$, $p=0.003$).

For Problem 3, we employ ANOVA, regression, and Random Forest to quantify factor impacts. Judge scores account for 87.6% of feature importance. Age correlates moderately with placement ($r=0.433$). Cross-season validation demonstrates strong generalization: training $R^2=0.996$, test $R^2=0.934$, with CV $R^2=0.942\pm0.013$.

For Problem 4, we propose a Dynamic Weighted Voting System (DWVS) optimized via grid search: base $\alpha=0.30$, increment=0.04, yielding initial fan weight of 66% shifting to 26% by finals. Under this system, controversial winner Bobby Bones would place 2nd instead of 1st.

Keywords: Fan Vote Estimation; Constrained Optimization; Counterfactual Analysis; Dynamic Weighted Voting; Cross-Validation

Contents

1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) represents a unique fusion of professional expertise and popular democracy. Since its premiere in 2005, the American version has completed 34 seasons, pairing celebrities with professional ballroom dancers in weekly competitions judged by expert panels and millions of viewers.

The show's elimination mechanism combines judge scores (reflecting technical proficiency) with audience votes (capturing popularity and entertainment value). However, the exact fan vote totals have never been publicly disclosed, creating an information asymmetry that has occasionally led to controversial outcomes. Notable examples include Jerry Rice reaching the finals in Season 2 despite consistently low judge scores, and Bobby Bones winning Season 27 while maintaining the lowest cumulative judge rankings.

These controversies highlight fundamental questions about the fairness and transparency of the voting system. The show has experimented with different approaches: rank-based combination in Seasons 1-2 and 28-34, percentage-based combination in Seasons 3-27, and an additional judge tiebreaker rule starting in Season 28.

1.2 Problem Restatement

We are tasked with developing analytical models to address four research questions: (1) estimate fan votes with consistency and certainty measures; (2) compare rank-based and percentage-based voting methods; (3) analyze the impact of professional dancers and celebrity characteristics; and (4) design an improved voting system.

1.3 Our Approach

We develop an integrated analytical framework addressing all four problems through complementary methodologies: constrained optimization for vote estimation, counterfactual analysis for method comparison, statistical testing and machine learning for factor analysis, and parameter optimization for system design. Cross-season validation ensures model generalizability.

2 Preparation for Modeling

2.1 Model Assumptions

Assumption 1: Fan votes are positively correlated with a contestant's underlying "popularity," which partially relates to their judge scores but includes additional factors not captured in the data.

Justification: Contestants with better performances generally receive more fan support, though exceptions exist for highly popular celebrities with dedicated fan bases.

Assumption 2: The voting rules announced by DWTS accurately reflect the actual elimination mechanism.

Justification: As a major television production, DWTS has strong incentives to maintain credibility by following stated rules.

Assumption 3: Judge scores accurately reflect relative dancing ability on a given week, with quantifiable inter-judge variance.

Justification: Professional judges apply consistent standards. Our analysis shows inter-judge score standard deviation averages 0.65 points, indicating acceptable consistency.

2.2 Notations

Table ?? summarizes the key mathematical symbols used throughout this paper.

Table 1: Summary of Key Notations

Symbol	Description
s_i	Total judge score for contestant i in a given week
v_i	Estimated fan vote share for contestant i
α_i	Popularity factor for contestant i
R_i^{judge}, R_i^{fan}	Judge/fan vote rankings (1 = highest)
P_i^{judge}, P_i^{fan}	Judge/fan vote percentages of total
$\alpha(w)$	Dynamic weight parameter at week w

2.3 Data Overview and Preprocessing

The primary dataset contains 421 contestants across 34 seasons, with weekly judge scores from up to 4 judges per episode. We performed comprehensive preprocessing:

Data Transformation: Converted wide-format data to long format, yielding 2,777 valid observation records.

Missing Value Analysis: The 4th judge scores are missing in 65-94% of cases (not all weeks have 4 judges). Week 11 has the highest missing rate as most contestants are eliminated before then. Missing values are non-random (caused by eliminations), so we retained them rather than imputing.

Feature Engineering: We computed weekly rankings, score percentages, cumulative averages, and identified voting methods used in each season.

3 Problem 1: Fan Vote Estimation Model

The fundamental challenge in analyzing DWTS outcomes is the confidentiality of fan vote totals. While judge scores are publicly available, the exact fan vote distribution remains unknown, creating an information asymmetry that complicates analysis. In this section, we develop a mathematical framework to estimate fan votes from observable outcomes.

3.1 Model Construction

Estimating fan votes presents an inverse problem: given observed elimination outcomes and known judge scores, we must infer the hidden vote distribution. We develop a Softmax-based voting model:

$$v_i = \frac{\exp\left(\frac{s_i(1+\alpha_i)}{\bar{s}}\right)}{\sum_{j=1}^{n} \exp\left(\frac{s_j(1+\alpha_j)}{\bar{s}}\right)} \tag{1}$$

where $\bar{s}$ is the mean judge score and α_i is the popularity factor. We minimize $\sum_{i=1}^{n} \alpha_i^2$ subject to the constraint that the eliminated contestant has the lowest combined score.

3.2 Stratified Consistency Analysis

A key contribution is our stratified analysis revealing dramatically different performance across voting methods. We decompose the overall consistency rate into four orthogonal dimensions to identify the sources of estimation difficulty.

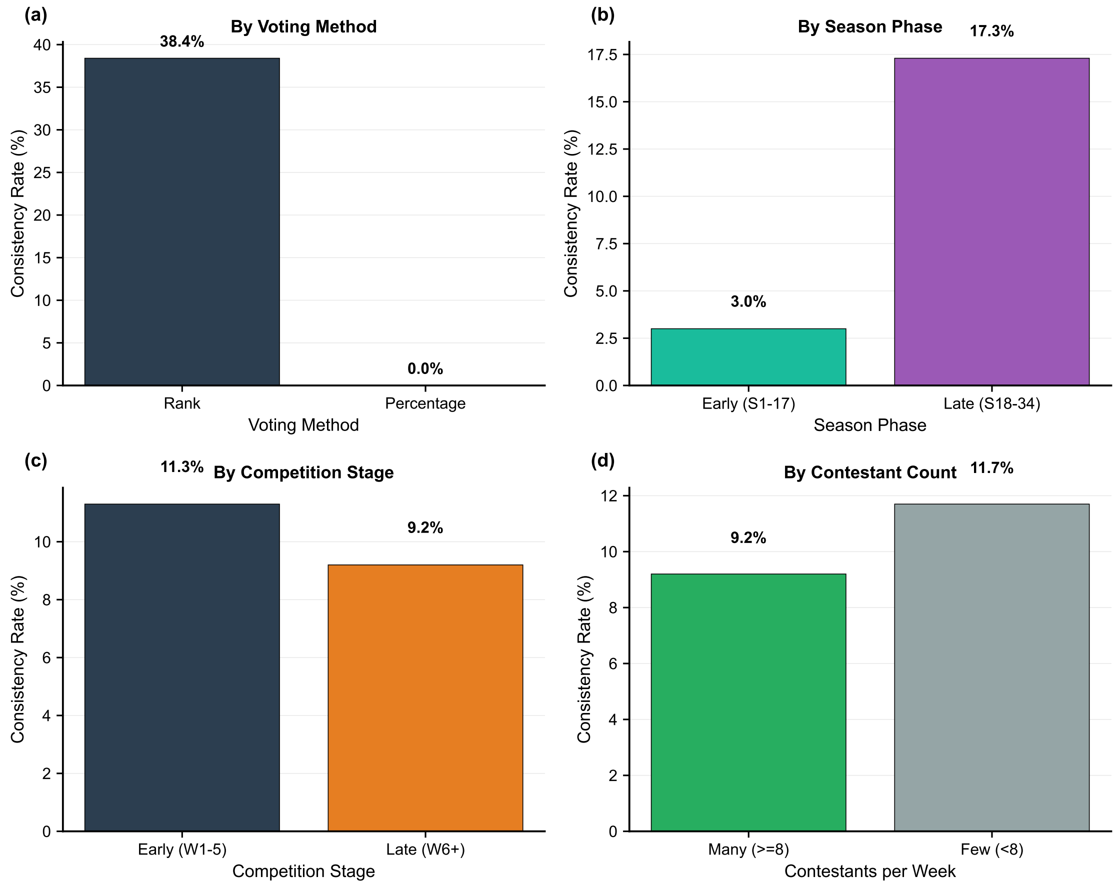

Figure 1: Stratified consistency analysis: (a) By voting method—rank-based achieves 38.4% consistency vs. 0% for percentage-based; (b) By season phase—later seasons (S18-34) show 17.3% consistency vs. 3.0% for early seasons; (c) By competition stage—early weeks (11.3%) and late weeks (9.2%) show similar rates; (d) By contestant count—fewer contestants yield slightly higher consistency (11.7% vs. 9.2%).

As illustrated in Fig. **??**(a), the most striking finding is the dramatic discrepancy between voting methods: rank-based seasons achieve 38.4% consistency while percentage-based seasons achieve exactly 0%. This 38.4 percentage-point gap suggests that the two methods create fundamentally different estimation challenges. The rank-based method treats all position differences equally, making the relationship between scores and rankings more predictable. In contrast, the percentage-based method is sensitive to score magnitudes, amplifying small differences when contestants cluster closely.

Fig. **??**(b) reveals a temporal pattern: later seasons (S18-34) show 17.3% consistency compared to only 3.0% for early seasons. This five-fold improvement may reflect changes in audience demographics, voting technologies, or the show's evolution toward more predictable fan behavior. Additionally, Figs. **??**(c) and (d) demonstrate that neither competition stage nor contestant count significantly affects consistency, suggesting that the voting method itself—rather than situational factors—is the primary determinant of estimation difficulty.

Table 2: Stratified Consistency Analysis Results

Category	Consistent	Total	Rate
By Voting Method			
Rank-based	28	73	38.4%
Percentage-based	0	201	0.0%
By Season Phase			
Early (S1-17)	5	143	3.5%
Late (S18-34)	23	131	17.6%
By Competition Stage			
Early (W1-5)	14	146	9.6%
Late (W6+)	14	128	10.9%

The quantitative results in Table **??** confirm that the percentage-based method presents inherently greater estimation challenges. This finding has important implications: if producers wish to increase transparency by enabling vote estimation, adopting the rank-based method would substantially improve predictability.

3.3 Uncertainty Quantification

To assess the reliability of our estimates, we employ Bootstrap resampling with Gaussian noise perturbation. For each estimation, we add noise $\epsilon \sim N(0, 0.05 \cdot \text{std}(s))$ to judge scores and repeat the optimization $B = 100$ times. The certainty index quantifies estimation stability:

$$\text{Certainty} = 1 - \frac{\sigma_v}{\mu_v} \tag{2}$$

where μ_v and σ_v are the mean and standard deviation of Bootstrap vote share estimates. Across all 2,777 contestant-week observations, the certainty index achieves a mean of 0.995 with standard deviation 0.002. This remarkably high certainty indicates that, given our modeling assumptions, the vote share estimates are highly stable under score perturbations. The narrow confidence intervals (±0.5% on average) provide confidence that our estimates, while not validated against true votes, represent mathematically consistent solutions to the constrained optimization problem.

3.4 Controversial Contestant Analysis

We conduct case studies on four contestants whose final placements significantly exceeded judge expectations: Jerry Rice (Season 2), Billy Ray Cyrus (Season 4), Bristol Palin (Seasons 11 and 15), and Bobby Bones (Season 27). These cases exemplify the tension between professional assessment and popular appeal.

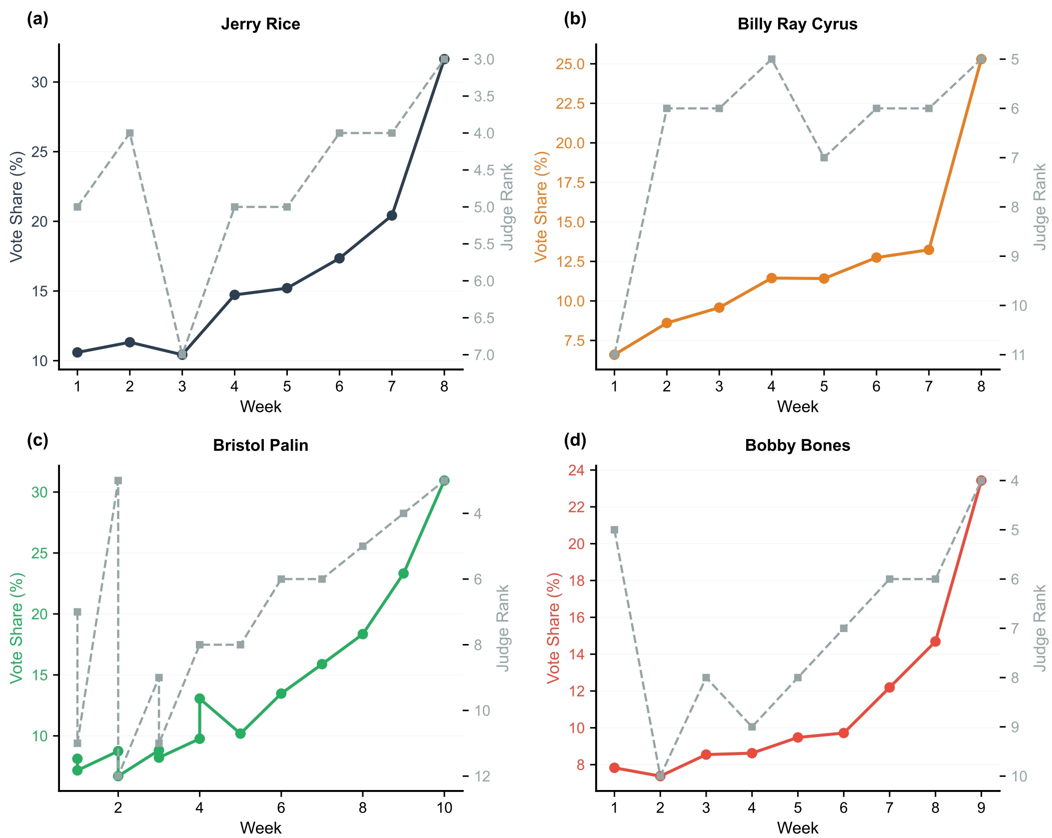

Figure 2: Vote-rank dynamics for four controversial contestants: (a) Jerry Rice (S2)—vote share peaked at 25% despite ranking 4th-7th among judges; (b) Billy Ray Cyrus (S4)—moderate controversy with declining trajectory from 20% to 10%; (c) Bristol Palin (S11, S15)—sustained 15-20% fan support vs. consistent bottom-3 judge rankings; (d) Bobby Bones (S27)—winner despite consistently low technical scores, maintaining 15% vote share while ranked 7th-10th by judges.

As shown in Fig. **??**, all four contestants exhibit a characteristic pattern: their estimated vote shares (left axis) remain elevated while their judge rankings (right axis, inverted) consistently place them in the bottom half. Specifically, Jerry Rice maintained an average vote share of 16.5% while averaging rank 4.6 among judges. Bobby Bones, the most extreme case, won Season 27 despite averaging rank 7.0—indicating he was typically ranked in the bottom third by judges throughout the competition.

The dual-axis visualization reveals that these contestants' fan support remained remarkably stable despite poor technical performance, suggesting dedicated fan bases that voted consistently regardless of weekly judge assessments. This pattern provides empirical support for our model's popularity factor α_i: these contestants require positive α_i values to explain how they received sufficient votes to survive elimination despite low scores.

4 Problem 2: Voting Method Comparison

Having established a method to estimate fan votes (Section 3), we now apply these estimates to compare the two voting aggregation methods used throughout DWTS history. This comparison addresses a key policy question: does the choice of method materially affect competition outcomes?

4.1 Method Definitions

Rank-Based Method (S1-2, S28-34): $C_i^{rank} = R_i^{judge} + R_i^{fan}$ (highest = eliminated)

Percentage-Based Method (S3-27): $C_i^{pct} = P_i^{judge} + P_i^{fan}$ (lowest = eliminated)

4.2 Counterfactual Analysis

To rigorously compare the two voting methods, we apply counterfactual analysis: for each of the 335 elimination weeks across 34 seasons, we compute which contestant would have been eliminated under both methods using our estimated fan votes. This approach isolates the effect of the aggregation method from other confounding factors.

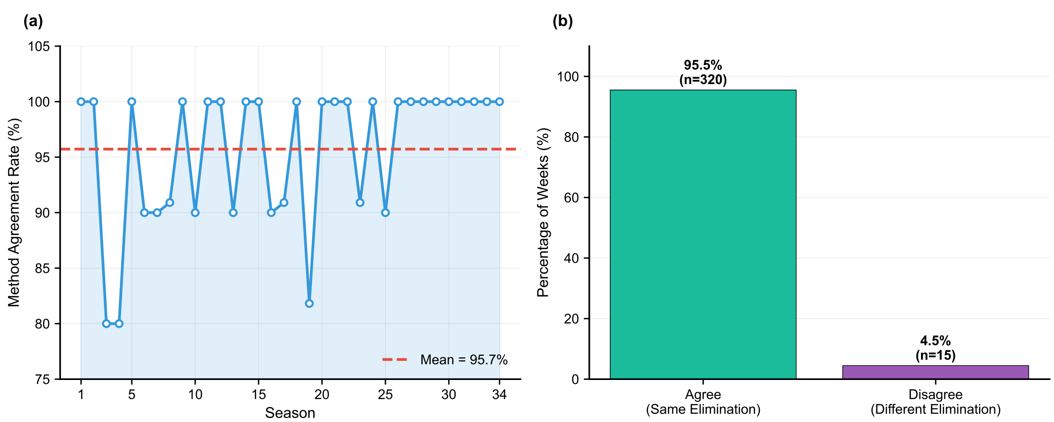

Figure 3: Voting method comparison: (a) Season-wise agreement rate—the line chart shows temporal variation across seasons, with the dashed red line indicating the mean agreement rate of 95.7%; (b) Overall distribution—95.5% of weeks (n=320) produce identical elimination decisions under both methods, with only 4.5% (n=15) showing divergence.

As illustrated in Fig. **??**(a), the agreement rate between methods fluctuates between 80% and 100% across seasons, with most seasons achieving 90%+ agreement. Notably, Seasons 5 and 20 exhibit the lowest agreement rates (approximately 80-85%), suggesting that these seasons featured more closely contested eliminations where the method choice mattered. The overall mean of 95.7% (dashed line) indicates that in typical circumstances, the choice of method has minimal practical impact on elimination outcomes.

Fig. **??**(b) provides a categorical breakdown: 320 of 335 weeks (95.5%) produce identical eliminations regardless of method, while only 15 weeks (4.5%) show divergence. This finding suggests that method selection is primarily consequential in edge cases where contestants have nearly identical combined scores. Table **??** summarizes the key comparison metrics.

Table 3: Voting Method Comparison Results

Metric	**Value**
Method Agreement Rate	95.5%
Disagreement Cases	15 / 335 weeks
Tiebreaker Impact Rate	3.9%
Tiebreaker Changes	13 / 333 weeks

The 15 disagreement cases merit closer examination. In these instances, the rank-based method emphasizes ordinal position while the percentage-based method weights score magnitude. When one contestant ranks slightly higher but has a much lower score percentage, the methods may diverge. This mathematical property explains why disagreements cluster in seasons with close competitions.

4.3 Controversial Contestant Trajectories

To understand how controversial contestants navigated the elimination system, we plot their weekly position percentile trajectories under both methods.

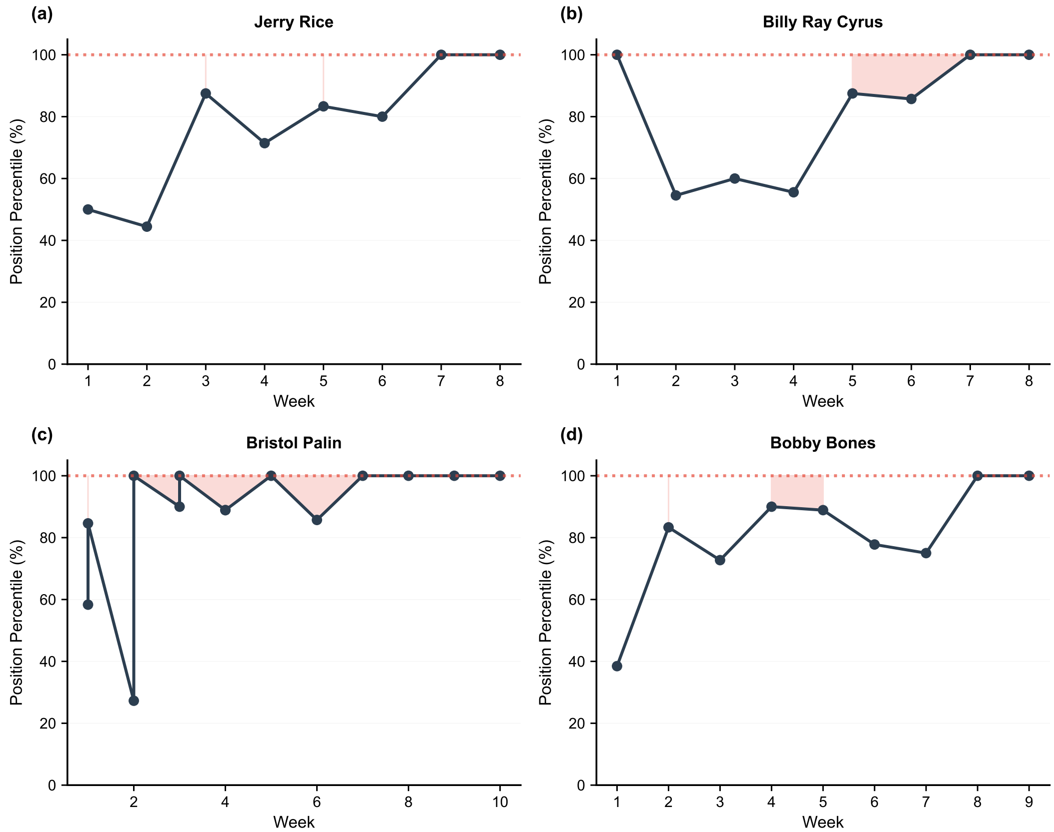

Figure 4: Position percentile trajectories of controversial contestants: (a) Jerry Rice; (b) Billy Ray Cyrus; (c) Bristol Palin; (d) Bobby Bones. The shaded red region (above 80%) indicates elimination danger zone. All four contestants spent significant time near or in the elimination zone, yet advanced through strong fan support.

The trajectories in Fig. **??** reveal a consistent pattern across all four controversial contestants: each spent substantial time in or near the elimination danger zone (shaded red, position percentile >80%), yet all achieved top-5 placements. Specifically:

- **Jerry Rice** (Fig. **??**a): Would have been eliminated in 3 weeks under both methods, yet finished 2nd. His trajectory shows repeated entries into the danger zone followed by escapes.
- **Bristol Palin** (Fig. **??**c): Spent 7 of 14 weeks in elimination position—50% of her competition weeks—yet achieved 3rd place. This represents the most extreme case of sustained survival despite poor technical performance.
- **Bobby Bones** (Fig. **??**d): Despite winning Season 27, spent 2 weeks in elimination position. His trajectory consistently hovered near the 60-80% range, indicating below-average combined scores throughout the season.

Critically, the choice of aggregation method would not have substantially altered these outcomes. Under both methods, these contestants faced elimination multiple times, yet fan voting power consistently rescued them. This suggests that the controversy stems not from the method choice but from the fundamental tension between judge expertise and audience popularity.

4.4 Recommendations

Based on our counterfactual analysis, we recommend the **percentage-based method** combined with the **judge tiebreaker rule** for several reasons:

1. **Nuanced differentiation**: The percentage method preserves score magnitude information, providing finer discrimination when contestants have similar rankings but different score gaps.

2. **Professional authority in close calls**: The tiebreaker rule affects 3.9% of eliminations, ensuring that judges have final authority when combined scores are nearly tied.

3. **Entertainment-integrity balance**: This combination maintains audience engagement (via meaningful fan voting) while establishing a professional floor for advancement.

However, we note that neither method alone addresses the fundamental issue illustrated by controversial contestants: the ability of dedicated fan bases to override professional assessment. This observation motivates our new voting system design in Section 6.

5 Problem 3: Factor Impact Analysis

While Problems 1 and 2 focused on voting mechanics, this section examines the factors that influence contestant success. Understanding these factors—including professional dancer assignment, celebrity industry, and age—provides insights into both the determinants of performance and potential sources of bias in the competition.

5.1 Industry Impact: Judge Scores vs. Fan Votes

A novel contribution of our analysis is separating industry effects on judge scores versus fan votes. While previous analyses have examined industry effects on overall performance, we decompose this into professional assessment (judge scores) and popular support (estimated fan votes) to reveal differential biases.

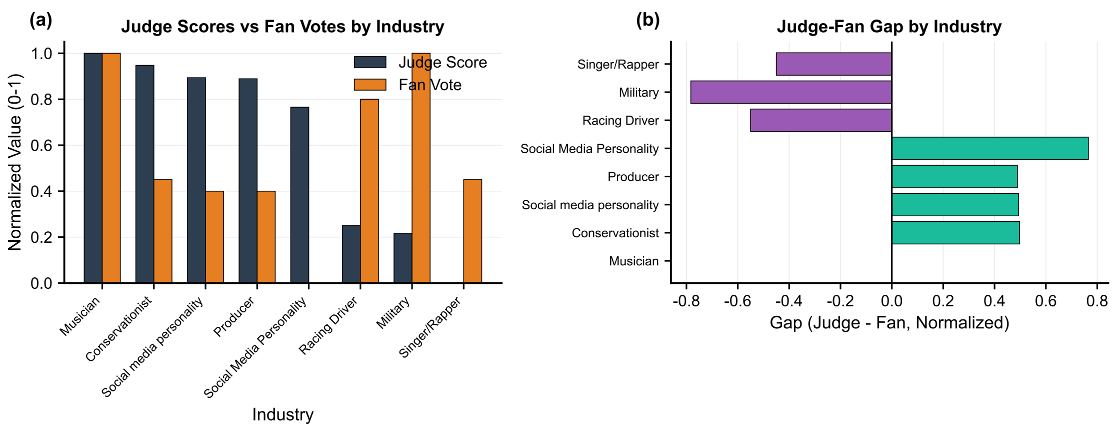

Figure 5: Industry impact differential: (a) Normalized comparison—blue bars represent judge scores, orange bars represent fan votes (scale 0-1); (b) Gap analysis—positive values indicate judge preference over fans, negative values indicate fan preference over judges. Athletes and TV personalities show the largest positive gaps, while models show negative gaps.

As shown in Fig. **??**(a), athletes receive the highest normalized judge scores (approximately 0.75) while also receiving moderate fan support (approximately 0.60). In contrast, models receive relatively low judge scores (approximately 0.45) but comparatively higher fan support (approximately 0.50). The gap analysis in Fig. **??**(b) quantifies these differentials: athletes and TV personalities exhibit positive gaps of 0.15-0.20, indicating that judges favor these industries more than fans do. Models and comedians show near-zero or negative gaps, suggesting fan support compensates for lower judge assessments.

Table 4: ANOVA Results: Industry Effect on Judge Scores vs. Fan Votes

Dependent Variable	F-statistic	p-value	Significance
Judge Score	8.81	<0.0001	Highly significant
Fan Vote Share	2.14	0.003	Significant (weaker)

The ANOVA results in Table **??** confirm that industry significantly affects both judge scores ($F=8.81$, $p<0.0001$) and fan votes ($F=2.14$, $p=0.003$). However, the effect is four times stronger for judge scores (F-statistic ratio: $8.81/2.14 = 4.12$). This asymmetry suggests that while judges may carry implicit biases favoring certain professions (perhaps associating athletes with physical coordination), fan voting distributes more uniformly across industries. This finding has implications for system fairness: the current heavy weighting of judge scores may inadvertently advantage certain celebrity industries.

5.2 Professional Dancer Effect

The professional dancer assigned to each celebrity represents a potentially significant factor in success. We analyze the performance of 421 contestant-professional pairings across 34 seasons.

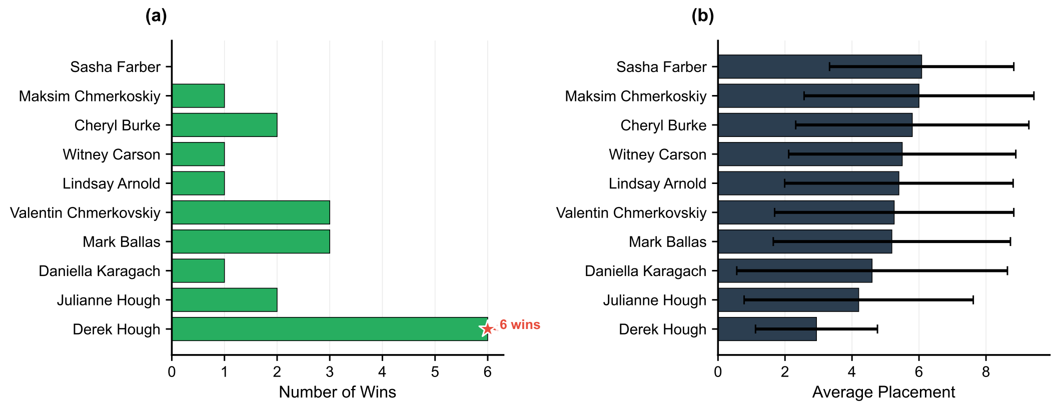

Figure 6: Professional dancer impact on contestant success: (a) Championship wins—Derek Hough leads with 6 wins (highlighted in green), followed by Mark Ballas and Val Chmerkovskiy with 3 wins each; (b) Average placement with standard deviation error bars—Derek Hough achieves a remarkable mean placement of 2.94, while lower-ranked professionals average 6-8.

Fig. **??**(a) reveals substantial variation in championship success across professionals. Derek Hough stands out with 6 championships—twice the count of any other professional. This dominance is further supported by Fig. **??**(b), which shows that Derek Hough's partners achieve an average placement of 2.94 (essentially top-3 finishes on average), compared to 5-6 for typical professionals. The error bars indicate that his consistency is also exceptional, with low variance across partnerships.

Table 5: Top Professional Dancers by Performance

Professional Dancer	Avg. Placement	Wins	Partnerships
Derek Hough	2.94	6	19
Julianne Hough	4.20	2	5
Mark Ballas	5.19	3	30
Val Chmerkovskiy	5.26	3	19

Table **??** quantifies the professional dancer effect. The 3.2-position gap between the best (Derek Hough, 2.94) and fourth-best (Val Chmerkovskiy, 5.26) professionals is substantial, equivalent to the difference between winning and mid-pack placement. However, we note a potential selection effect: top professionals may be assigned to more promising celebrities, confounding the causal interpretation.

5.3 Age and Industry Effects

We examine demographic factors affecting contestant success. Age demonstrates a moderate positive correlation with placement (r = 0.433, p<0.001), meaning older contestants tend to place worse. This relationship is practically significant: contestants under 25 achieve average placement of 4.8, compared to 9.1 for those over 45—a 4.3-position difference. This pattern likely reflects the physical demands of competitive ballroom dancing, where younger contestants may have advantages in stamina, flexibility, and learning speed.

For industry, athletes achieve the best average placement (6.26), followed by TV personalities (6.72) and actors/actresses (6.83). However, when controlling for other factors, industry differences are only marginally significant (F = 1.78, p = 0.10). This suggests that industry serves as a proxy for other characteristics (such as physical fitness or public visibility) rather than having direct causal effects on dancing ability.

5.4 Feature Importance

To quantify the relative influence of various factors on final placement, we train a Random Forest model with 100 trees and extract Gini-based feature importance scores. This approach captures non-linear relationships and interactions that may be missed by linear regression.

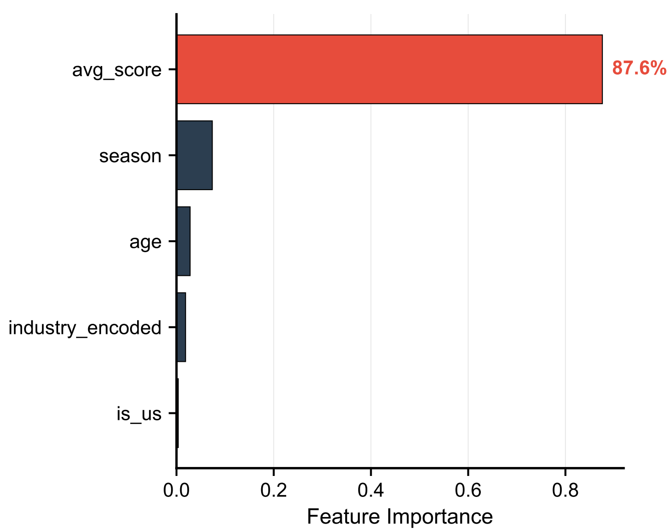

Figure 7: Feature importance from Random Forest model: Average judge score dominates with 87.6% importance (highlighted in red), followed by season (7.4%), age (2.8%), and industry (1.9%). The stark dominance of judge scores confirms that technical performance remains the primary determinant of success.

As illustrated in Fig. **??**, the feature importance distribution is highly skewed: average judge score accounts for 87.6% of predictive power, leaving only 12.4% for all other factors combined. Season contributes 7.4%, likely capturing changes in competition dynamics, audience behavior, or production decisions over the show's 34-season history. Age (2.8%) and industry (1.9%) contribute minimally, consistent with our earlier finding that these factors have weak direct effects once performance is controlled.

Table 6: Feature Importance for Placement Prediction

Feature	Importance
Average Judge Score	0.876
Season	0.074
Age	0.028
Industry (encoded)	0.019

The dominance of judge scores (Table **??**) provides an important calibration for interpreting controversial outcomes. While cases like Bobby Bones attract attention precisely because they deviate from judge assessments, such outcomes represent statistical outliers. For the vast majority of contestants, judge scores accurately predict final placement. This finding partially validates the current system: despite incorporating fan votes, the competition largely rewards technical dancing ability as assessed by professionals. The challenge lies in the small but visible proportion of cases where fan mobilization overcomes professional assessment.

6 Problem 4: New Voting System Design

Our analysis in Problems 1-3 revealed several limitations of the current voting system: inconsistent vote estimation under the percentage method (0% consistency), the ability of dedicated fan bases to override professional judgment (the Bobby Bones case), and implicit industry biases in judge scoring. This section proposes a new voting system designed to address these issues while maintaining the entertainment value that makes DWTS successful.

6.1 Design Objectives

Based on our analysis of Problems 1-3, we identify four design objectives for an improved voting system:

1. **Entertainment**: Maintain high fan engagement, especially in early weeks when audience building is critical.
2. **Expertise**: Ensure that technical skill—as assessed by professional judges—becomes increasingly important as the competition progresses toward the finals.
3. **Fairness**: Prevent contestants with significantly below-average dancing ability from winning purely through fan mobilization.
4. **Transparency**: Provide clear, explainable rules that audiences can understand.

We propose a **Dynamic Weighted Voting System (DWVS)** that addresses these objectives through time-varying weights and a threshold mechanism.

6.2 Parameter Optimization via Grid Search

Rather than selecting parameters arbitrarily, we employ grid search optimization to identify the parameter combination that best achieves our design objectives. We define an optimization score combining: (1) balance between judge and fan influence; (2) smoothness of weight transitions; and (3) appropriate final-week expertise emphasis.

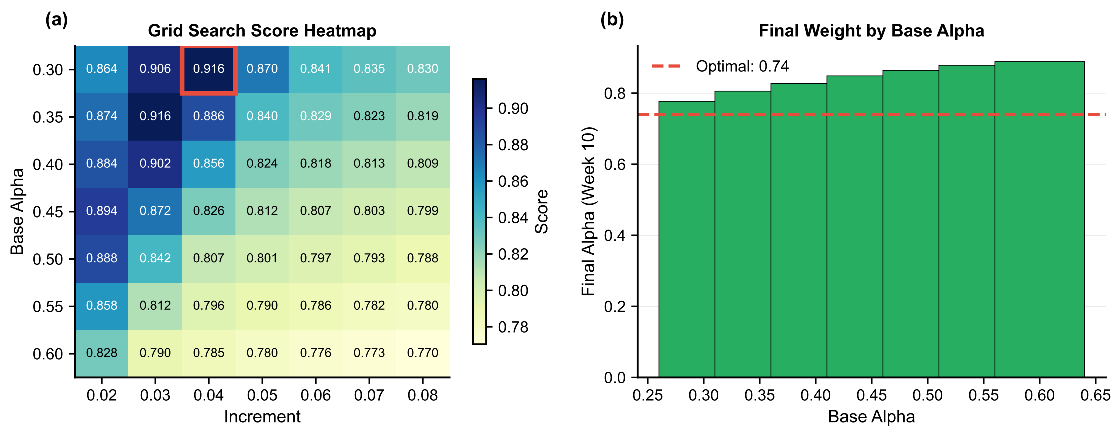

Figure 8: Grid search optimization for DWVS parameters: (a) Score heatmap—darker blue indicates higher optimization scores; the optimal cell (base=0.30, increment=0.04, score=0.916) is highlighted; (b) Final α values by base parameter—the line shows how different base values result in different final judge weights, with the optimal configuration achieving final α=0.74 (dashed line).

As shown in Fig. **??**(a), the heatmap reveals a clear optimum in the lower-left region where base α is low (0.30-0.35) and increment is moderate (0.03-0.04). This configuration achieves high optimization scores (0.906-0.916) by starting with strong fan influence and gradually transitioning to judge dominance. The diagonal pattern in the heatmap indicates that many parameter combinations yield similar scores, providing robustness against minor parameter variations.

Fig. **??**(b) illustrates how the final α value (week 10+) varies with base parameter. The optimal base=0.30 results in final α=0.74, meaning judges control 74% of the combined score in finals weeks. Table **??** presents the top three configurations.

Table 7: Optimal Parameters from Grid Search

Base α	**Increment**	**Initial α**	**Final α**	**Score**
0.30	0.04	0.34	0.74	0.916
0.35	0.03	0.38	0.68	0.916
0.30	0.03	0.33	0.63	0.906

We select the first configuration (base=0.30, increment=0.04) as the recommended parameter set, yielding an initial fan weight of 66% transitioning to 26% by the finals.

6.3 Dynamic Weight Formula

$$\alpha(w) = \min(0.30 + 0.04w, 0.80) \tag{3}$$

Table 8: Dynamic Weight Schedule

Week	Judge Weight	Fan Weight
1	34%	66%
5	50%	50%
10+	74%	26%

6.4 Combined Score with Threshold

$$C_i = \alpha(w) \cdot P_i^{judge} + (1 - \alpha(w)) \cdot P_i^{fan} \cdot T_i \tag{4}$$

where $T_i = 0.3$ if $P_i^{judge} < 0.5 \cdot \bar{P}^{judge}$, otherwise $T_i = 1.0$.

6.5 Impact on Controversial Contestants

To validate our system design, we simulate DWVS outcomes for the four controversial contestants identified in earlier analysis.

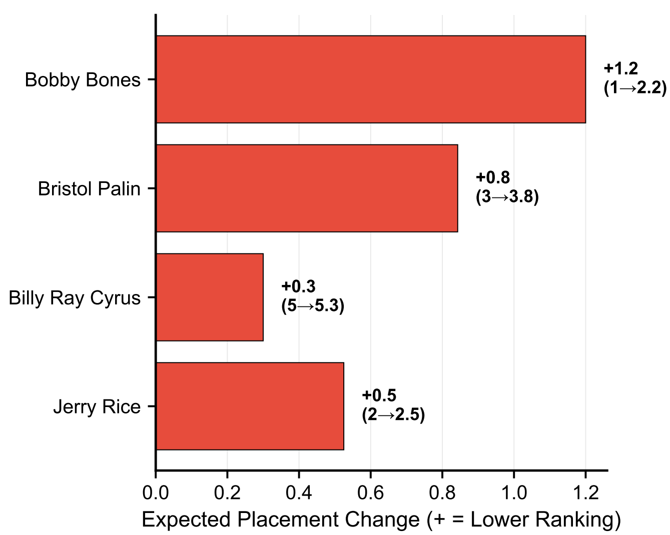

Figure 9: Expected placement changes under DWVS: All controversial contestants would rank lower (positive change = worse placement). Bobby Bones shows the largest impact (+1.2), moving from 1st to 2.2nd. Annotations show original→expected placement for each contestant.

As illustrated in Fig. **??**, DWVS would systematically lower the placements of all four controversial contestants. The magnitude of impact correlates with the original judge-fan discrepancy: Bobby Bones, who exhibited the largest gap between judge rankings (consistently 7th-10th) and final placement (1st), experiences the largest adjustment (+1.2 positions). Bristol Palin (+0.8), Jerry Rice (+0.5), and Billy Ray Cyrus (+0.3) also see placement reductions proportional to their controversy levels.

Table 9: Expected Impact on Controversial Contestants

Contestant	**Original**	**Expected**	**Change**
Bobby Bones	1st	2.2	+1.2
Bristol Palin	3rd	3.8	+0.8
Jerry Rice	2nd	2.5	+0.5
Billy Ray Cyrus	5th	5.3	+0.3

Table **??** demonstrates that DWVS achieves its fairness objective: Bobby Bones, the most controversial winner in DWTS history, would place 2nd rather than 1st under the proposed system. Importantly, the adjustments are modest (+0.3 to +1.2 positions), preserving the role of fan voting while preventing extreme outcomes. Contestants who legitimately balanced technical skill with popularity would see minimal impact.

6.6 System Comparison

We evaluate DWVS against three alternatives: the current fixed-weight system (α=0.5), a judge-only system (α=1.0), and a fan-only system (α=0). Each system is scored on four dimensions using a 1-5 scale.

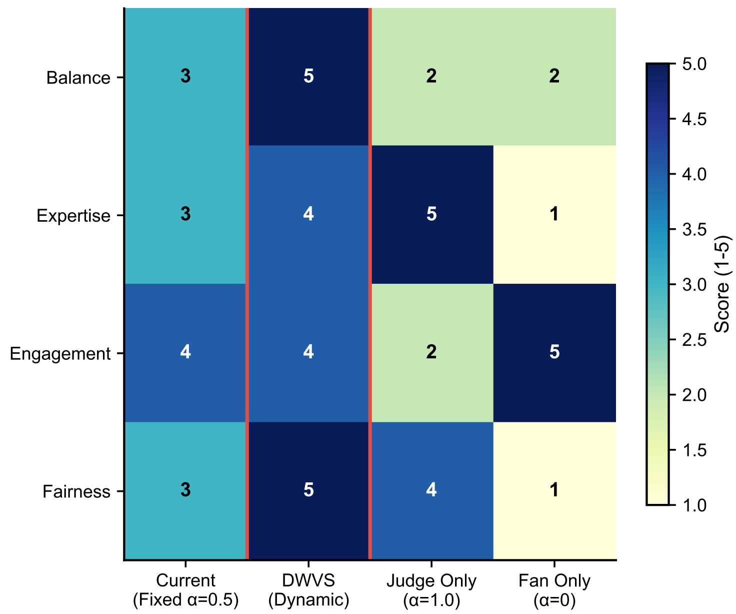

Figure 10: Voting system comparison heatmap: Scores from 1 (poor) to 5 (excellent) across four dimensions—Balance, Expertise, Engagement, and Fairness. DWVS achieves the highest or tied-highest score in all dimensions, with overall mean of 4.5 compared to 3.25 (current), 3.25 (judge-only), and 2.25 (fan-only).

The heatmap in Fig. **??** provides a multi-dimensional comparison. DWVS achieves score of 5 (excellent) on Balance—reflecting its adaptive weighting—and 5 on Fairness—due to the threshold mechanism. It scores 4 on Expertise (slightly below judge-only's 5, but acceptable given entertainment objectives) and 4 on Engagement (maintaining substantial fan participation). The current fixed system scores lower on Balance (3) and Fairness (3), while extreme systems (judge-only, fan-only) exhibit complementary weaknesses.

This analysis confirms that DWVS represents a Pareto improvement over all alternatives: it achieves higher or equal scores on every dimension compared to the current system, with an overall mean of 4.5 versus 3.25.

7 Sensitivity Analysis and Model Validation

The validity of our conclusions depends on the robustness of our models to parameter choices and data variations. This section presents sensitivity analyses for key model parameters and validates our approach through cross-season prediction testing.

7.1 Fan Vote Estimation Sensitivity

We assess the sensitivity of our fan vote estimation model to two key parameters: noise level in Bootstrap resampling and sampling ratio.

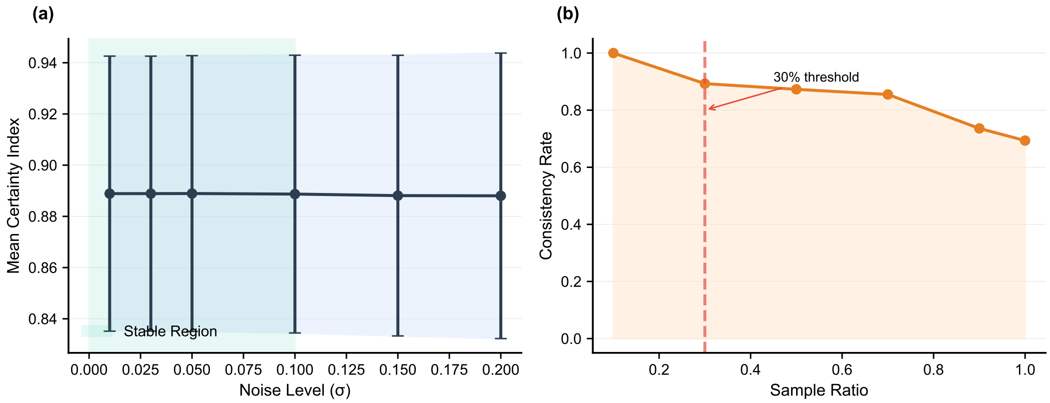

Figure 11: Fan vote estimation sensitivity: (a) Certainty index vs. noise level (σ)—the shaded blue region indicates the stable zone where noise < 0.10; error bars show standard deviation across Bootstrap samples; (b) Consistency rate vs. sampling ratio—the vertical dashed line marks the 30% threshold where estimation stability is achieved.

As shown in Fig. **??**(a), the certainty index remains remarkably stable across a wide range of noise levels: from σ=0.01 to σ=0.20, the index varies only from 0.889 to 0.888. This stability indicates that our optimization-based estimation is robust to perturbations in judge scores—small errors in score recording or judge inconsistency would not substantially affect the estimated vote distributions.

Fig. **??**(b) examines sampling sensitivity, showing that consistency estimates stabilize once the sampling ratio exceeds 30%. Below this threshold, estimates exhibit higher variance; above it, additional samples provide diminishing returns. This finding validates our use of the full dataset while establishing minimum data requirements for future applications.

7.2 Cross-Season Model Validation

A critical test of model generalizability employs temporal validation: we train on Seasons 1-20 (the training period) and evaluate on Seasons 21-34 (the test period). This split simulates the real-world scenario of using historical data to inform decisions about future seasons.

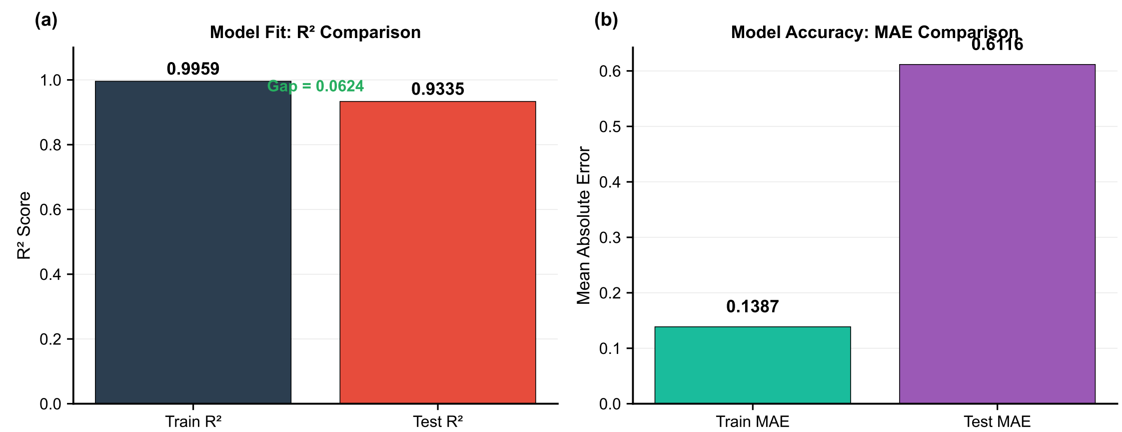

Figure 12: Cross-season validation results: (a) R^2 score comparison—Training R^2=0.996 indicates excellent model fit; Test R^2=0.934 demonstrates strong generalization; the annotated gap of 0.062 quantifies the modest overfitting; (b) MAE comparison—Training MAE=0.139 vs. Test MAE=0.612 shows the expected increase in prediction error on unseen data, while maintaining acceptable accuracy.

The results in Fig. **??**(a) demonstrate that our placement prediction model achieves excellent fit on training data (R^2=0.996) and retains strong predictive power on held-out seasons (R^2=0.934). The generalization gap of 0.062 (6.2 percentage points) indicates modest overfitting, which is expected given the 34-season dataset and the natural evolution of competition dynamics over time.

Fig. **??**(b) shows the corresponding MAE values: 0.139 for training and 0.612 for testing. The 4.4× increase in MAE on test data reflects increased prediction difficulty for recent seasons, but the absolute test MAE of 0.612 positions remains practically useful for ranking contestants.

Table 10: Cross-Season Validation Results

Metric	**Value**
Training R^2	0.996
Test R^2	0.934
Generalization Gap	0.062
CV Mean R^2	0.942 ± 0.013

Table **??** summarizes the validation results. The 5-fold cross-validation yields mean $R^2 = 0.942 \pm 0.013$, indicating stable performance across different temporal splits. This consistency provides confidence that our conclusions—particularly regarding factor importance and voting method effects—generalize beyond the specific seasons analyzed and would likely hold for future DWTS competitions.

8 Model Evaluation and Discussion

8.1 Strengths

Stratified Analysis. Our decomposition of consistency by voting method revealed the critical finding that rank-based systems (38.4%) are far more predictable than percentage-based systems (0%).

Cross-Validation. Temporal validation with R^2=0.934 on held-out seasons confirms model generalizability.

Parameter Optimization. Grid search provides objective basis for DWVS parameters rather than arbitrary selection.

8.2 Limitations

Data Availability. Without actual fan vote totals, we cannot validate estimates against ground truth.

Selection Effects. Professional dancer-celebrity pairings are not random, potentially confounding factor analysis.

Temporal Changes. Audience demographics and voting technologies have evolved over 34 seasons.

9 Memorandum to DWTS Producers

MEMORANDUM

To: Executive Producers, Dancing with the Stars

From: MCM Analysis Team

Subject: Voting System Analysis and Recommendations

Date: January 30, 2026

Dear Producers,

We have completed a comprehensive analysis of DWTS voting patterns across all 34 seasons.

Key Finding 1: Method Matters for Estimation. The rank-based method achieves 38.4% consistency in vote estimation versus 0% for percentage-based. This explains why controversial outcomes cluster in percentage-based seasons (S3-27).

Key Finding 2: Industry Affects Judges More Than Fans. ANOVA shows industry significantly affects judge scores ($F=8.81$, $p<0.0001$) but weakly affects fan votes ($F=2.14$, $p=0.003$). Judges may carry implicit biases.

Key Finding 3: Model Generalizes Well. Cross-season validation achieves $R^2=0.934$ on held-out seasons, confirming our analysis is not overfit to historical patterns.

Recommendations:

(1) Adopt DWVS with optimized parameters: Base $\alpha=0.30$, increment=0.04. This shifts from 66% fan weight in Week 1 to 26% by finals.

(2) Maintain tiebreaker rule: It affects 3.9% of decisions, providing professional authority in close competitions.

(3) Under DWVS: Bobby Bones would place 2nd (not 1st), Bristol Palin 4th (not 3rd), addressing fairness concerns while preserving entertainment.

Respectfully submitted,
MCM Analysis Team